Phage Propagation Liquid Amplification (Large-Scale Lysate) Protocol

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To amplify bacteriophage using Xanthomonas host bacteria and generate high-titer phage lysate. Used to detect phage and confirm infection. Used to expand phage stock.

Materials

- Log-phase Xanthomonas culture ($OD_{600} \approx 0.4–0.6$)
- Phage stock
- NYG soft agar (0.6–0.7%)
- NYG agar plates
- NYG liquid medium
- 30°C incubator

Procedure

1. Grow 50 mL Xanthomonas to OD600 ~0.4.
2. Add phage at MOI ~0.01–0.1.
    a. MOI (Multiplicity of Infection) is calculated by dividing the total number of phage particles added by the total number of bacterial cells present.
    total phage = phage titer (PFU/mL) × volume added (mL).
    total bacteria = cell concentration (cells/mL) × culture volume (mL).
    MOI = total phages ÷ total bacteria.
3. Incubate at 30°C, 200 rpm.
4. Monitor culture:
    - Culture becomes clear/turbid drop → lysis occurring.
    - Typically 4–8 hours.
5. After complete lysis:
    - Centrifuge at 4000 x g for 10 min to remove debris.
    - Collect supernatant (contains phage).
6. Filter sterilize (0.22 µm filter).

This is crude phage lysate.

Phage Storage

- Store lysate at 4°C (short-term).
- Add chloroform (few drops) if long-term bacterial contamination risk.
- Avoid freezing unless protocol validated.

Explore the Notebook